JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION
SPRDL1-24-R-0122

1. **Contracting Agency:** Defense Logistics Agency - Land Warren (DLA-WRN).

2. **Description of Action:** DLA proposes to award a new Firm-Fixed-Price contract to RENK AMERICA LLC Commercial and Government Entity (CAGE) 02978, TECMOTIV USA, INC., (CAGE),0L7R7, or an offeror providing the product of the approved source(s) for National Stock Number's (NSN) 2815-01-233-9709 and 2815-01-422-4201, Crankshaft Assembly and Connecting Rods, using Army Working Capital Funds.

3. **Description of Supplies/Services:** The supplies to be acquired under the proposed contract are Crankshaft Assemblies and Connecting Rods, which are used on the M88 Family of Vehicles (FOV). The total estimated contract value, with option, is [redacted]

Purchase Request Order Number	EH4C0427EH	EH4C0526EH
Item Name	Crankshaft Assembly	Connecting Rod
NSN	2815-01-233-9709	2815-01-422-4201
Manufacturer Part Number's	12254249: 02978 388010002: 0L7R7	E9AR104-0290: 02978 388010006:0L7R7
Base Quantity	62 each	228 each
Option Quantity	62 each	228 each
Estimated Unit Price	[redacted]	[redacted]
Estimated Value	[redacted]	[redacted]
Estimated Option Value	[redacted]	[redacted]
Estimated Total	[redacted]	[redacted]

EH4C0427EH:

Acquisition Method Code (AMC):
1 - Suitable for competitive acquisition for the second or subsequent time.

Acquisition Method Suffix Code (AMSC):
C - This part requires engineering source approval by the design control activity in order to maintain the quality of the part. An alternate source must qualify in accordance with the design control activity's procedures, as approved by the cognizant Government engineering activity.

[redacted]

EH4C0526EH:

Acquisition Method Code (AMC):
1 - Suitable for competitive acquisition for the second or subsequent time.

Acquisition Method Suffix Code (AMSC):
R- The Government does not own the data or the rights to the data needed to purchase this part from additional sources. It has been determined to be uneconomical to buy the data or rights to the data. It is uneconomical to reverse engineer the part. This code is used when the Government did not initially purchase the data and/or rights. If only one source has the rights or data to manufacture this item, AMCs 3, 4, or 5 are valid. If two or more sources have the rights or data to manufacture this item, AMCs1 or 2 are valid.

4. Authority Cited: The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204(a)(1) as implemented by Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. Reason for Authority Cited:
The Air Cooled, V-engine configuration, Diesel, Super turbocharged (AVDS)-1790 family of diesel engines are the heart of the M88 Recovery Vehicle Platform that make mobility and recovery operations possible. The crankshaft assemblies are an integral part of the AVDS-1790 family of diesel engines. The crankshaft is heart of the AVDS-1790 engine's rotating assembly. It is geometrically tailored to fit within the engine block architecture. The crankshaft has specific bearing journal spacing and diameters so that it can fit securely into the block's main caps that secure it. Also, the connecting rod journal size is uniquely sized to the crankshaft rod journals and the crankshaft offset, or 'throw' has been chosen to accommodate the stroke and valve clearance dimensional needs.

The connecting rods are in turn attached to the crankshaft journals. There are 12 connecting rods within the AVDS-1790 8CR engine. The connecting rods are steel I-beam forgings which 'connect' the piston to the crankshaft assembly and transfer the reciprocating motion generated by the piston to rotational motion of the crankshaft. During the combustion cycle, the forces imparted onto the piston are then transferred through the connecting rods and transferred to the crankshaft assembly giving the AVDS-1790 8CR its power output capabilities.

The M88's primary role is to extricate combat/tactical vehicles that have been damaged, stuck, overturned, or inoperable, as well as the vehicles crew. Recovery missions can take place during training exercises or in a combat zone. The M88's secondary role is to aide in the repair or replacement of damaged parts during maintenance operations. The M88 is commonly used to remove/install engines and/or transmissions for the Abrams FOV, Bradley FOV, other M88 FOV and M113 Armored Personnel Carrier.

[redacted]

Connecting Rod:
The Government does not have the pertinent technical data to enable competitive re-procurement. The Government does not own the necessary intellectual property data rights to the pertinent technical data to enable competitive re-procurement.

Crankshaft Assembly:
The Government has a build to print Technical Data Package (TDP) and/or a full specification that it can build to. However, the Government requires qualification prior to source approval.

Both approved sources have proven the ability to manufacture crankshafts and connecting rods that can meet the MIL-DTL-62177 durability specification which is an adapted form of the North Atlantic Treaty Organization (NATO) requirement for successful completion of NATO Laboratory Test for Gas Turbine Engines Allied Engineering Publication (AEP)-5, while operating on an AVDS-1790 family of diesel engines. This test is required for components which are critical for engine/vehicle operation and crew safety. [redacted] The approved sources have proven that they are able to deliver connecting rods and crankshafts of sufficient quality in support of the AVDS-1790 family of diesel engines. If another possible source of supply for this item were identified, qualification testing would be required.

A complete TDP is fully developed to support a competitive procurement for the crankshafts however, due to the necessary manufacturing expertise and low return on expenditure for a source qualification effort, the Government is not actively pursuing new sources for the crankshaft, assembly. A fully developed, build-to-print TDP for the connecting rods is not available. Only Renk America LLC. and Tecmotiv (USA), Inc. have the experience and capability to manufacture connecting rods that can withstand the [redacted] NATO endurance test within an AVDS-1790 8CR Engine. A combined reverse engineering effort to develop a fully mature TDP is estimated to cost [redacted] and would require [redacted] months to develop. Additionally, testing of the reverse engineered connecting rods and qualification of an additional source for the crankshaft would require successful completion of the [redacted] NATO and vehicle level durability testing required to prove out the technical data. The required vehicle level Reliability, Availability, and Maintainability (RAM) testing is estimated at an additional cost of [redacted] and will take approximately [redacted] to complete without any major issues. These estimated costs were derived from historical vehicle testing efforts and similar reverse engineering efforts within the U.S. Government. A cost estimate breakdown is as follows:

Reverse Engineering Effort:

Concurrent:	
Acquisition of (quantity of two) new crankshafts:	$[redacted]
Obtain sample set of quantity of 12 connecting rods:	$[redacted]
Acquisition of (quantity of two) AVDS-1790 8CR test assets:	[redacted]
Follow-on:	
Engine teardown and rebuild with prototype crank and rods:	$[redacted]
Reverse Engineer & technical data development:	$[redacted]
Prototyping, Integration, Dyno-Test:	$[redacted]
Final Qual. (NATO 400):	$[redacted]
Sub Total: $2.9M ~31 months	
Vehicle Level Testing RAM Testing of 5,000 miles:	$[redacted]
GRAND TOTAL:	$[redacted]

There are currently no plans to qualify additional manufacturing sources for the crankshaft assemblies or connecting rods. The Government would not be able to recover, through subsequent competition, the duplication of costs involved in procuring the data or reverse engineering to increase competition in a reasonable amount of time.

It is estimated that 10% savings could be achieved by adding an additional supplier. Considering the current average usage rates and the collective yearly cost of $[redacted] spent on the connecting rods and crankshaft assemblies a 10% saving achieved through competition would yield a potential [redacted] in annual savings. This would take the Government approximately [redacted] years to recoup the estimated $[redacted] cost of qualifying another source.

Based on the latest information provided by the U.S Army Tank-automotive and Armaments Command (TACOM) Procurement Coordinator the current stock on hand (SOH) for the crankshaft assemblies is [redacted] The current SOH for the connecting rods is [redacted] for the connecting rods with the AMD being [redacted]. [redacted]

6. Efforts to Obtain Competition:

a. Effective competition: Due to the reasons cited in Section 5, the Government is not pursuing competition for this item. Notices required by FAR 5.201 will be posted.

b. Subcontracting competition: Unless the contract is awarded to a small business, the solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in

Subcontracting"; 52.219-16, "Liquidated Damages--Subcontracting Plan";" and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)," to maximize small business opportunities and utilization.

7. **Actions to Increase Competition:** The Government would not be able to recover the $[redacted] to qualify a new manufacturing source of supply within a justifiable timeframe needed to support the demand that this J&A is intended to address. Therefore, the Government is not pursuing a new source qualification effort at this time.

8. **Market Research:** A Sources Sought Notice was posted to SAM.gov on and was open for 30 days. One response was received. BHPE LLC (CAGE: 9SWS7) has provided a response as an interested vendor. Engineering was not able to assess BHPE's capability as an interested vendor as BHPE did not provide any documentation regarding their capabilities to produce this item. The response received was generic and didn't demonstrate their ability to produce the Crankshaft Assemblies and Connecting Rods that could be considered for use by the Government.

Market research was performed by U.S Army Combat Capabilities Development Command (DEVCOM) Sustainment Engineering on 21 May 2024 by reviewing vendor on-line catalogs and available product literature for the crankshafts published by Scat Crankshafts Inc. and Ohio Crankshaft Company to determine if commercial items or non-developmental items are available to meet Government's needs or could be modified to meet the Government's needs. While these two sources have the ability to manufacture crankshafts and crankshaft assemblies that could theoretically be used as solutions in the M88 AVDS-1790 engines, their existing crankshafts lack the unique individualized design which is unique to the AVDS-1790 air-cooled diesel application. Additionally, neither Scat Crankshafts Inc. nor Ohio Crankshaft Company currently manufacture a crankshaft designed with the precise overall length, journal spacing, and dimensions needed for this particular application. Finally, since neither of these sources have produced this specific item for the M88 they have not completed the necessary testing to become a qualified source

For the connecting rods the market research was performed by DEVCOM Heavy Combat Sustainment Engineering on 20 May 2024 by reviewing on-line vendor catalogs and available product literature published by CP-Carrilo Inc., and Scat Enterprises Inc. to determine if commercial items or non-developmental items are available to meet Government's needs or could be modified to meet the Government's needs. While these two sources have the ability to manufacture connecting rods for diesel applications that could theoretically be used as solutions in the M88 AVDS engines, their existing connecting rods are not specifically tailored to the crankshaft journal diameter and are not the required length unique to the AVDS-1790 8CR air-cooled diesel application. Finally, since neither of these sources have produced this specific item for the M88, they have not completed the necessary testing to become a qualified source.

Keyword: Heavy Duty Engine Crankshaft Manufacturers+USA

Website: https://www.ohio-crankshaft.com/products/crankshafts/
Findings: No drop-in solution available to meet the AVDS-1790 Diesel engines block architecture. A clean-sheet engineering effort would be required.

Keyword: Connecting Rod Manufacturer for Diesel Engines

Website: https://www.cp-carrillo.com/pt-3508-rods.html
Findings: No solutions found specific to AVDS-1790. One suitable for the AVDS-1790 would need to be developed.

Website: https://www.scatcrankshafts.com/product-category/connecting-rods/?wpf_fbv=1
Findings: No solutions found specific to AVDS-1790. One suitable for the AVDS-1790 would need to be developed.

Website: https://www.scatcrankshafts.com/product-category/crankshafts/?wpf_fbv=1
Findings: No drop-in solution available to meet the AVDS-1790 Diesel engines block architecture. A clean-sheet engineering effort would be required.

9. Interested Sources: To date no additional sources have expressed interest beyond the one listed above in Section 8.

10. Other Facts: Procurement History:

CRANKSHAFT ASSEMBLY

1) Contract Number: SPRDL1-24-C-0012
Award Date: 02 November 2023
Contractor: EOS INC (CAGE; 85VY7) (Approved Distributor-Current)
Contract Value: $3,097,400
Competitive Status: FAR 6.302-1(a)(2), Limited Source

2) Contract Number: SPRDL1-20-C-0030
Award Date: 15 November 2019
Contractor: BRIGHTON CROMWELL (CAGE; 3NNX8) (Approved Distributor-Former)
Contract Value: $3,411,213.60
Competitive Status: FAR 6.302-1(a)(2), Limited Source

3) Contract Number: SPRDL1-16-C-0174
Award Date: 13 April 2016
Contractor: L3 TECHNOLOGIES (CAGE; 02978) (Subsequently acquired by Renk)
Contract Value: $534,270

Competitive Status: FAR 6.302-1 only one responsible source

4) Contract Number: W56HZV-08-C-0463
Award Date19 September 2008
Contractor: L3 TECHNOLOGIES (CAGE; 02978) (Subsequently acquired by Renk)
Contract Value: $2,803,206.42
Competitive Status: FAR 6.302-1 only one responsible source

CONNECTING ROD, PISTON

1) Contract Number: SPRDL1-22-C-0065
Award Date: 5 September 2022
Contractor: TECMOTIV USA (CAGE; 0L7R7)
Contract Value: $1,486,499.20
Competitive Status: FAR 6.302-1(a)(2), Limited Source

2) Contract Number: SPRDL1-18-C-0194
Award Date: 26 March 2018
Contractor: BRIGHTON CROMWELL (CAGE; 3NNX8) (Approved Distributor)
Contract Value: $408,040.22
Competitive Status: FAR 6.302-1(a)(2), Limited Source

3) Contract Number: SPRDL1-17-P-0184
Award Date: 18 April 2017
Contractor: BRIGHTON CROMWELL (CAGE; 3NNX8) (Approved Distributor)
Contract Value: $118,610.70
Competitive Status: FAR 6.302-1(a)(2), Limited Source

4) Contract Number: SPRDL1-16-C-0121
Award Date: 04 February 2016
Contractor: L3 TECHNOLOGIES (CAGE; 02978) (Subsequently acquired by Renk)
Contract Value: $181,944
Competitive Status: FAR 6.302-1 only one responsible source

5) Contract Number: SPRDL1-13-C-0121
Award Date 09 July 2016
Contractor: L3 TECHNOLOGIES (CAGE; 02978) (Subsequently acquired by Renk)
Contract Value: $173,123.96
Competitive Status: FAR 6.302-1 only one responsible source

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

[redacted]

Name: [redacted]
Title: Division Chief Sustainment Engineering
Signature:

10/2/2024

X [redacted]

Signed by: USA

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [redacted]
Title: Division Chief – Sourcing Division
Signature:

9/24/2024

X [redacted]

Signed by: [redacted]

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This determination will be based on the results of a detailed cost and/or price analysis. The contractor will be required to submit certified cost or pricing data as this item has been determined to be other than a commercial product.

Name: Jacob Silcox
Title: Contracting Officer
Signature:

9/24/2024

X Jacob Silcox

Signed by: SILCOX.JACOB.A.1368798920

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Jacob Silcox
Title: Contracting Officer
Signature:

9/24/2024

X Jacob Silcox

Signed by: SILCOX.JACOB.A.1368798920

[redacted]

[redacted]

Approval Authority for Acquisitions Greater than $750,000 But Less Than $15M

Based on the foregoing Justification, I hereby approve the following procurement of the following:

a. Noun: Crankshaft, Assembly and Connecting Rods

b. NSN: 2815-01-233-9709, 2815-01-422-4201

c. Basic Quantity: 62 Each, 228 Each

d. Basic Estimated Value: $[redacted], [redacted]

e. Option Percentage: 100%, 100%

f. Quantity with Option: 124 Each, 456 Each

g. Total Estimated Maximum Value with Option: $[redacted]

This action will be awarded under other than full and open competition basis pursuant to the authority of 10 U.S.C. 3204(a)(1) and subject to the availability of funds. This approval is subject to the provision that the property herein described has otherwise been authorized acquisition.

Signature:

WIEBUSCH.JAMIE.[redacted]
X[redacted]

Digitally signed by WIEBUSCH.JAMIE[redacted]
Date: 2024.10.09 11:34:15 -04'00'

Competition Advocate

[redacted]